Lecture 3 Scribe

L3 (1).pdf
PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 3 slides).Instructions:Use 'One-Shot Prompting'. I will provide you with an Example (The Shot) of how to convert a slide concept into a scribe format. You must follow this structure and style to generate the scribe for the Lecture 3 slides.EXAMPLE (THE SHOT):Input Context: 'Slide 1: Definition of Set. A set is a collection of distinct objects. Example: A = {1, 2, 3}. Notation: x ∈ A means x is in set A.'Output Scribe:'### 1. Fundamentals of SetsDefinition: A set is a well-defined collection of distinct objects.Notation:$A = \{1, 2, 3\}$ represents a set with elements 1, 2, and 3.$x \in A$ denotes that element x belongs to set A.'YOUR TASK:Now, generate the full lecture scribe for the attached Lecture 3 slides following the style and faithfulness of the example above.Faithfulness: Do not add outside information. Use exact definitions/notations from slides.Structure: Use clear headings (e.g., Sample Space, Events, Probability Axioms) as per the slides.Goal: Exam-ready reference material."

Got it — I've generated an **exam-ready lecture scribe** for the provided *Lecture 3* slide deck, strictly following your one-shot format (concise headings + definitions/bullets derived only from the slides, no outside material). Since this particular Lecture 3 deck primarily covers **course orientation, structure, and project framework (not probability theory yet)**, the scribe reflects exactly that content.

Lecture 3 — CSE400: Fundamentals of Probability in Computing

(Introduction, Course Structure, and Project Framework)

1. Course Overview

Course Title: CSE400 – Fundamentals of Probability in Computing
Lecture: 3
Instructor: Dr. Dhaval Patel, PhD (Associate Professor, SEAS–Ahmedabad University)
Research Lab: MICxN (Machine Intelligence, Computing and xG Networks)
Lecture Date: January 13, 2026

This lecture introduces the course ecosystem, learning philosophy, engagement methods, and the semester-long project structure.

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2. Learning Mindset

Two contrasting mindsets are highlighted:

Growth Mindset (emphasized):

- Failure is an opportunity to grow.
- Challenges help growth.
- Effort and attitude determine abilities.
- Feedback is constructive.
- Inspired by others' success.

Fixed Mindset (contrasted):

- Failure reflects limits of ability.
- Abilities are unchanging.
- Avoids challenges.
- Gives up when frustrated.

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3. Outline and Motivation

General Course Information

- Team (Instructor + TAs)
- Active Learning Platform: **Campuswire**
- Schedule
- Grading

Why Learn CSE400?

- Daily life conversations as motivating examples.

Engineering Applications

- Speech Recognition
- System Radar Systems
- Communication Networks

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4. Course Team

Instructor

- Dr. Dhaval Patel
- Role: Instructor
- Office: Faculty Office (Room-210)
- Areas of Interest: xG Networks, Applied ML/DL/RL/AutoML, Intelligent Transportation Systems, Life Sciences, Behaviour Modelling using AI
- Email: dhaval.patel@ahduni.edu.in

Teaching Assistants / Student Team

- Deep Patel – RL and Pinching Antenna Systems
- Prapti Patel – Smart Sensing Frameworks for 5G/Beyond
- Raj Koticha – Multi-Agent RL for Resource Management
- Ritu Patel – Intelligent Transportation Systems
- Rushi Moliya – UAV Deployment Optimization
- Ura Modi – Pinching Antenna Systems

(Each listed with academic year, project focus, and university email.)

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5. Active Learning and Communication

Course Website

Two Campuswire sections are provided for enrolled students.

Why Campuswire?

- Anonymous participation to build confidence.
- Collaborative and active learning.
- Real-time feedback via polling.
- Direct messaging with instructor/TAs.

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6. Schedule and Interaction

Lecture Sessions

- **Section-1:** 9:30 AM – 11:00 AM (Tuesday, Thursday), GICT Room-136
- **Section-2:** 1:00 PM – 2:30 PM (Tuesday, Thursday), GICT Room-137

TA Hours

- In person / online (timings to be announced).

Instructor Interaction

- 24×7 contact via Campuswire.
- Best practice: post queries on Campuswire.
- Direct messages allowed for private discussions.
- External engagement options: UGRP-8 (2026), offline projects, counseling, or short meetings (via email).

Important Note:

- LaTeX tutorial and assignment submission guidelines are emphasized.

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7. Evaluation Component: Project (30%)

Project Weightage: 30%

Team Formation

- Deadline: 17 January 2026 (Saturday, End of Day)

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8. Project Execution Framework

Six major milestones (M1–M6):

- **M1:** Kickstart – Team formation, area identification, background, motivation, problem formulation.
- **M2:** Mathematical Modeling – RVs, PMF/PDF, CDF, multivariate RVs, joint PMF/PDF/CDF.
- **M3:** Coding – Simulation + computation.
- **M4:** Inference – Choose, understand, and code a randomized algorithm.
- **M5:** Randomized Algorithms – Apply to domain problem and compare with deterministic approaches.
- **M6:** Bounds + Analysis – Final compilation and submission.

9. Project Deliverables and Assessment

Major Artifacts

- Concept Evolution Maps
- Scribe: Process & Decision-Making (decision logs, constraints, alternatives, evidence, trade-off matrices)
- Multimodal Artifacts (video/audio/visual)
- Question-Driven Artifacts
- Collaboration & Team Dynamics Artifacts

Deliverables: Codes, reports, videos (as specified).

Assessment: Before and after mid-semester.

Final Evaluation: Project viva + submission.

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10. Submissions

Submission #1 – Concept Evolution Maps

Tools:

- Miro
- diagrams.net (draw.io)

Submission #2 – Scribe (Learning Reflection Logs)

Types:

- Lecture Scribes
- Project Scribe

Key Points:

- Two groups per lecture generate scribes.
- Must reflect lecture content with examples from foreign textbooks.
- Minimum 8–10 pages.
- Total 6 bi-weekly submissions.

Submission #3 – Multimodal Artifacts

- One video per milestone (~10–15 minutes).
- Explain milestone + coding/simulation (if applicable).
- Think-aloud videos / one-minute insight videos / project demos.

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11. Undergraduate Research Programme (UGRP)

UGRP philosophy emphasizes:

- Multidisciplinary learning (Arts, Science, Management)
- Experiential and research-driven education
- 4D Model: Discover → Design → Develop → Deliver
- T-shaped engineer concept (depth + breadth)

The lecture also showcases:

- MICxN lab activities
- Industry collaborations
- International academic partners
- Alumni outcomes
- Conference publications and awards
- Case studies of previous UGRP scholars and their career trajectories.

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12. Closing

The lecture concludes with an open Q&A session and contact information for continued engagement.

Source: Lecture 3 slide deck provided by the instructor

If you'd like, next we can convert later lectures (once probability theory formally begins) into the same exam-ready scribe format.

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 3 slides).Instructions:Use 'One-Shot Prompting'. I will provide you with an Example (The Shot) of how to convert a slide concept into a scribe format. You must follow this structure and style to

generate the scribe for the Lecture 3 slides.EXAMPLE (THE SHOT):Input Context: 'Slide 1: Definition of Set. A set is a collection of distinct objects. Example: A = {1, 2, 3}. Notation: x ∈ A means x is in set A.'Output Scribe:'### 1. Fundamentals of SetsDefinition: A set is a well-defined collection of distinct objects.Notation:$A = \{1, 2, 3\}$ represents a set with elements 1, 2, and 3.$x \in A$ denotes that element x belongs to set A.'YOUR TASK:Now, generate the full lecture scribe for the attached Lecture 3 slides following the style and faithfulness of the example above.Faithfulness: Do not add outside information. Use exact definitions/notations from slides.Structure: Use clear headings (e.g., Sample Space, Events, Probability Axioms) as per the slides.Goal: Exam-ready reference material."

Understood — below is the **Lecture 3 scribe** written strictly in the *one-shot style* you provided (compact headings + definition/bullet expansion), using **only** what appears in the uploaded slides. Note that this Lecture 3 deck focuses on **course orientation, learning framework, and project structure** (not probability concepts yet), so the scribe reflects exactly that.

Lecture 3 — CSE400: Fundamentals of Probability in Computing

1. Course Introduction

Course: **CSE400 – Fundamentals of Probability in Computing**
Lecture: **3 (Introduction to Probability Theory – course orientation session)**
Instructor: **Dr. Dhaval Patel, PhD (Associate Professor)**
Affiliation: SEAS–Ahmedabad University, MICxN Research Lab
Date: January 13, 2026

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2. Learning Mindset

Growth Mindset:

- Failure is an opportunity to grow.
- I like to try new things.

- I can learn to do anything I want.
- Challenges help me grow.
- My effort and attitude determine my abilities.
- Feedback is constructive.
- I am inspired by the success of others.

Fixed Mindset:

- Failure is the limit of my abilities.
- I'm either good at it or I'm not.
- My abilities are unchanging.
- I don't like to be challenged.
- My potential is predetermined.
- When I'm frustrated, I give up.
- I stick to what I know.

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3. Course Outline

CSE400: General Course Information

- Team
- Active Learning Platform: **Campuswire**
- Schedule
- Grading

Why should we learn CSE400?

- Example: Daily life conversations

Engineering Applications

- Speech Recognition
- System Radar System
- Communication Network

---

4. Course Team

Instructor:

- Dr. Dhaval Patel
- Role: Instructor
- Place: Faculty Office (Room-210)

- Areas of Interest: xG Networks, Applied ML/DL/RL/AutoML, Intelligent Transportation Systems, Life Sciences, Behaviour Modelling using AI
- Email: dhaval.patel@ahduni.edu.in

Student Team (TAs / Project Members):

- Deep Patel – RL and Pinching Antenna Systems
- Prapti Patel – Smart Sensing Frameworks for 5G and Beyond
- Raj Koticha – Multi-Agent Reinforcement Learning for Resource Management
- Ritu Patel – Intelligent Transportation System
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(Each listed with degree year, project topic, and university email.)

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5. Active Learning and Class Discussion

Course Website (Campuswire Sections)

- Section-1 link
- Section-2 link

Why use Campuswire?

- Build confidence: anonymous participation and post/back-channel communication.
- Collaborative and active learning: solve difficulties with online participation.
- Feedback on lecture session: real-time polling.
- Personal communication with instructor/TAs: direct messaging.

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6. Schedule

Lecture Sessions

- Section-1: 9:30 AM – 11:00 AM (Tuesday, Thursday), GICT Room-136
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TA Hours

- In person / online (timings to be finalized).

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7. Connecting with the Instructor

- Contact hours: 24×7 through Campuswire.
- Best practice: post queries on Campuswire.
- Direct messages allowed for private student–instructor discussions.
- External engagement: UGRP-8 (2026), offline projects, counseling, or short meetings (via office email).

Important Note:

- LaTeX tutorial and assignment submission guidelines.

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8. Project Component

Project Kickoff
Project Weightage: **30%**

Team Formation
Deadline: **17 January 2026 (Saturday, End of Day)**

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9. Project Execution Guidelines

Major Milestones (M1–M6):

- **M1 – Kickstart:** Team formation, area identification, background, motivation, problem formulation.
- **M2 – Mathematical Modeling:** RV, PMF/PDF, CDF, multivariate RVs, joint PMF/PDF/CDF.
- **M3 – Coding:** Simulation + computation.
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- **M5 – Randomized Algorithms:** Apply randomized algorithms to the domain problem and compare results with deterministic algorithms.
- **M6 – Derive Bounds + Analysis:** Compile and submit final deliverables.

Deliverables: Codes, reports, videos, etc. (as specified).
Assessment: Before and after mid-semester.
Final Evaluation: Project viva and submission.

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10. Submissions

Submission #1 – Concept Evolution Maps

Tools:

- Miro
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Submission #2 – Scribe: Learning Reflection Logs
Types:
(a) Lecture Scribes
(b) Project Scribe

Key points:

- Two groups per lecture generate scribes.
- Minimum 8–10 pages.
- Includes process & decision-making (decision logs, constraints, alternatives, evidence, trade-off matrices).
- Total 6 submissions in bi-weekly mode.

Submission #3 – Multimodal Artifacts

- One video per milestone (~10–15 minutes).
- Explain milestone and coding/simulation (if applicable).
- Think-aloud videos / one-minute insight videos / project demos.

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11. Undergraduate Research Programme (UGRP)

Core ideas presented:

- Multidisciplinary learning (Arts, Science, Management).
- Experiential, research-driven approach.
- 4D Model: **Discover + Design + Develop + Deliver.**
- T-shaped individual: depth in one discipline and breadth across disciplines.

Slides also present MICxN lab activities, industry projects, international collaborations, alumni outcomes, conference publications, awards, and UGRP scholar case studies.

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12. Conclusion

Lecture ends with an open Q&A session and contact information for further discussion.

Source: Lecture 3 slide deck (uploaded PDF)

If you'd like, you can upload the **next lecture (where Sample Space / Events / Probability Axioms start)** and I'll convert that into the same exam-focused scribe format.

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Speech Recognition.
System Radar System.
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Raj Koticha – Proactive Self-Aware Multi-Agent Reinforcement Learning for Resource Management.
Ritu Patel – Working in Intelligent Transportation System.
Rushi Moliya – Optimizing UAV Deployment and Multi-UAV 3D Deployment.
Ura Modi – Working in Pinching Antenna Systems.

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5. Active Learning and Class Discussion

Course Website: Campuswire (two sections provided).

Why use Campuswire?
Build confidence: anonymous participation and post/back-channel communication.
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Project Weightage: 30%.

Project Team Formation:
Deadline: 17 January 2026 (Saturday, End of Day).

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Includes process & decision-making: decision logs (Why X over Y?), constraints, alternatives considered, final decision, evidence used, trade-off matrices (cost vs. performance vs. risk).
Total 6 bi-weekly submissions.

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One video per milestone (~10–15 minutes).
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Think-aloud videos / one-minute insight videos / project demo.

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Emphasis on:
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4D Model: **Discover + Design + Develop + Deliver.**
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12. Conclusion

Lecture concludes with an open Q&A session and instructor contact information.

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